

Chapter 8: M-codes

8.1 Introduction

This chapter gives detailed descriptions of the M-codes that you use to program your machine.

8.1.1 List of M-codes



CAUTION:

The sample programs in this manual have been tested for accuracy, but they are for illustrative purposes only. The programs do not define tools, offsets, or materials. They do not describe workholding or other fixturing. If you choose to run a sample program on your machine, do so in Graphics mode. Always follow safe machining practices when you run an unfamiliar program.



NOTE:

The sample programs in this manual represent a very conservative programming style. The samples are intended to demonstrate safe and reliable programs, and they are not necessarily the fastest or most efficient way to operate a machine. The sample programs use G-codes that you might choose not to use in more efficient programs.

M-codes are miscellaneous machine commands that do not command axis motion. The format for an M-code is the letter M followed by two to three digits; for example M03.

Only one M-code is allowed per line of code. All M-codes take effect at the end of the block.

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M00 Stop Program

The M00 code stops a program. It stops the axes, spindle, and turns off the coolant (including optional Through Spindle Coolant, Through Tool Air Blast, and Auto Air Gun / Minimum Quantity Lubrication). The next block after the M00 is highlighted when viewed in the program editor. Press **[CYCLE START]** to continue program operation from the highlighted block.

M01 Optional Program Stop

M01 works the same as M00, except the optional stop feature must be on. Press **[OPTION STOP]** to toggle the feature on and off.

M02 Program End

M02 ends a program.



NOTE:

The most common way of ending a program is with an M30.

M03 / M04 / M05 Spindle On Fwd/On Rev/Stop

M03 turns spindle on in the forward direction. M04 turns spindle on in the reverse direction. M05 stops the spindle. For spindle speed, refer to G96/G97/G50.

M08 Coolant On / M09 Coolant Off

M08 turns on the optional coolant supply and M09 turns it off. For High Pressure Coolant, see M88/M89.

M10 Chuck Clamp / M11 Unclamp

M10 clamps the chuck and M11 unclamps it.

The direction of clamping is controlled by Setting 282 (refer to page400 for more information).

M12 / M13 Auto Jet Air Blast On/Off (Optional)

M12 and M13 activate the optional Auto Air Jet. M12 turns the air blast on and M13 turns the air blast off. M12 Srrr Pnnn (rrr in RPM and nnn is in milliseconds) turns the air blast on for the specified time, rotates the spindle at the specified speed while the air blast is on, then turns off both the spindle and the air blast automatically. The air blast command for the secondary spindle is M112/M113.

M14 / M15 Main Spindle Brake On/Off (Optional C-Axis)

These M Codes are used for machines equipped with the optional C-Axis. M14 applies a caliper-style brake to hold the main spindle, while M15 releases the brake.

M17 / M18 Turret Rotation Fwd/Rev

M17 and M18 rotate the turret in the forward (M17) or reverse (M18) direction when a tool change is made. The following M17 program code causes the tool turret to move forward to tool 1 or reverse to tool 1 if an M18 is commanded.

```
N1 T0101 M17 (Forward) ;
```

```
N1 T0101 M18 (Reverse) ;
```

An M17 or M18 stays in effect for the remainder of the program.



NOTE:

Setting 97, Tool Change Direction, must be set to M17/M18.

M19 Orient Spindle (Optional)

M19 adjusts the spindle to a fixed position. The spindle only orients to the zero position without the optional M19 orient spindle feature.

The orient spindle function allows P and R address codes. For example, M19 P270 . orients the spindle to 270 degrees. The R value allows the programmer to specify up to two decimal places; for example, M19 R123.45. View the angle in the **Current Commands Tool Load** screen.

M119 positions the secondary spindle (DS lathes) the same way.

Spindle orientation is dependent on the mass, diameter, and length of the workpiece and/or the workholding (chuck). Contact the Haas Applications Department if any unusually heavy, large diameter, or long configuration is used.

M21 / M22 Tailstock Advance/Retract (Optional)

M21 and M22 position the tailstock. M21 uses Settings 341 and 342 to move to the Tailstock Advance Distance. M22 uses Setting 105 to move the tailstock to the Retract Point.



NOTE:

ST10 does not use any settings (105, 341, 342).

Adjust pressure using the valves on the HPU (except ST-40, which uses Setting 241 to define hold pressure). For information on proper ST tailstock pressure, refer to pages **108** and **109**.

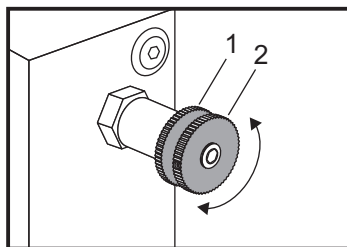


CAUTION:

Do not use an M21 in the program if the tailstock is positioned manually. If this is done, the tailstock backs away from the workpiece and then repositions against the workpiece, which may cause the workpiece to drop.

F8.1:

Set Screw Hold Pressure Valve: [1] Locking knob, [2] Adjustment knob.



M23 / M24 Chamfer Out of Thread On/Off

M23 commands the control to execute a chamfer at the end of a thread executed by G76 or G92. M24 commands the control not to perform chamfering at the end of the threading cycles (G76 or G92). An M23 remains in effect until changed by M24, likewise for M24. Refer to Settings 95 and 96 to control the chamfer size and angle. M23 is the default at power-up and when the control is reset.

M30 Program End and Reset

M30 stops a program. It stops the spindle and turns off the coolant and the program cursor returns to the start of the program.



NOTE:

M30 no longer cancels tool length offsets.

M31 / M33 Chip Auger Forward/Stop (Optional)

M31 starts the optional chip auger motor in the forward direction (the direction that moves the chips out of the machine). The auger does not turn if the door is open. It is recommended that the chip auger be used intermittently. Continuous operation causes the motor to overheat. Settings 114 and 115 control the auger duty cycle times.

M33 stops auger motion.

M35 Parts Catcher Mid-Point Position

M35 allows the parts catcher to move to the middle position.

This allows for easier setup and time savings instead of fully retracting the parts catcher, you can command M35 to move the parts catcher out of the way.

This function has been added to the Parts Catcher device page. To access the page press the **[CURRENT COMMANDS]** button then go to the Devices tab.

M36 / M37 Parts Catcher On/Off (Optional)

M36 rotates the parts catcher into position to catch a part. M37 rotates the parts catcher out of the work envelope.

M38 / M39 Spindle Speed Variation On/Off

Spindle Speed Variation (SSV) allows the operator to specify a range within which the spindle speed continuously varies. This is helpful in suppressing tool chatter, which can lead to an undesirable part finish and/or damage to the cutting tool. The control varies the spindle speed based on Settings 165 and 166. For example, in order to vary spindle speed +/- 50 RPM from its current commanded speed with a duty cycle of 3 seconds, set Setting 165 to 50 and Setting 166 to 30. Using these settings, the following program varies the spindle speed between 950 and 1050 RPM after the M38 command.

M38/39 Program Example

```
%  
o60381 (M38/39-SSV-SPINDLE SPEED VARIATION) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
S1000 M3 (Turn spindle CW at 1000 RPM) ;  
G04 P3. (Dwell for 3 seconds) ;  
M38 (SSV ON) ;  
G04 P60. (Dwell for 60 seconds) ;  
M39 (SSV OFF) ;  
G04 P5. (Dwell for 5 seconds) ;  
G00 G53 X0 (X home) ;  
G53 Z0 (Z home & C unwind) ;  
M30 (End program) ;  
%
```

The spindle speed continuously varies with a duty cycle of 3 seconds until an M39 command is found. At that point the machine comes back to its commanded speed and the SSV mode is turned off.

A program stop command such as M30 or pressing **[RESET]** also turns SSV Off. If the RPM swing is larger than the commanded speed value, any negative RPM values (below zero) translates into an equivalent positive value. The spindle, however, is not allowed to go below 10 RPM when SSV mode is active.

Constant Surface Speed: When Constant Surface Speed (G96) is activated (which calculates spindle speed) the M38 command alters that value using Settings 165 and 166.

Threading Operations: G92, G76 and G32 allow the spindle speed to vary in SSV mode. This is not recommended due to possible thread lead errors caused by mismatched acceleration of the spindle and the Z-axis.

Tapping cycles: G84, G184, G194, G195, and G196 are executed at their commanded speed and SSV is not applied.

M41 / M42 Low/High Gear (Optional)

On machines with a transmission, M41 selects low gear and M42 selects high gear.

M43 / M44 Turret Unlock/Lock (Service Use Only)

For Service use only.

M51-M58 Set Optional User M-Codes

The M51 through M58 codes are optional for user interfaces. They activate one of the relays and leave it active. Use M61-M68 to turn these off. **[RESET]** turns off all of these relays.

Refer to M121 through M128 on page 360 for details on the M-code relays.

M59 Set Output Relay

P - Discrete output relay from 0 to 255 or macro number 12000 to 12255.

M59 turns on a discrete output relay. An example of its usage is M59 Pnnn, where nnn is the relay number being turned on. M59 can also be used with the corresponding macro number in the range from 12000 to 12255. When using Macros, M59 P90 does the same thing as using the optional macro command #12090=1, except that it is processed at the end of the line of code.



NOTE:

For the 8 spare M functions on relay board 1 use relays 90-97 or macro addresses #12090 - #12097